

Group 7 Status Report

Management of Organizational Data (YR 25-26)

➤ Overview

When the pandemic hit in early 2020, the sudden shift to remote work and online learning disrupted consumer behavior. Overnight, kitchen tables became classrooms and home offices, driving unexpected changes in demand for traditional office supplies.

Our group is exploring how this transformation unfolded at the local level: how long it took foot traffic to recover, and whether dwell times indicated changes in consumer intent.

➤ Industry, Geographic Region or Metropolitan Focus

Our analysis centers on the Office Supply Store category within the United States. We analyze trends at the county level. The office supply sector provides a valuable case study for understanding how remote work adoption and family composition influenced local economic recovery during the pandemic.

➤ Initial Research and Specific Questions to answer

1. How did foot traffic and dwell times at office supply stores evolve week by week during the first four months of 2020, and which demographic factors, including education levels, working-age population (18-64), and K-12 (5-17) households suggest about potential hot spots for innovative concept stores or expanded community services?
2. Did suburban counties experience a stronger rise in visits and dwell time at office supply stores than urban centers during the first wave of remote work?
3. Did certain office supply brands experience faster stabilization in visits and dwell times than others during the first four months of 2020, and what might this suggest about brand loyalty and essential purchasing behavior?

➤ Progress So Far

- Queried weekly visitation and dwell-time trends for office supply store brands.
- Conducted exploratory analyses to visualize weekly declines and rebounds in visits.
- Identified where office supply stores are most concentrated
- Looked at the top states with highest traffic for office supply category
- Looked at total number of records and date coverage

➤ Next Steps

- Model demographic predictors for visit and dwell time rebound using regression.
- Deepen fine-grained temporal analyses (by week and by brand).
- Systematically compare suburban and urban recovery in the context of local restrictions.
- Integrate more policy data to explain outlier cases.

Query Log

```
-- Get dataset size and verify time coverage for NAICS 453210
```

```
SELECT
  COUNT(*) AS total_rows,
  MIN(v.date_range_start) AS start_date,
  MAX(v.date_range_start) AS end_date,
  COUNT(DISTINCT v.date_range_start) AS distinct_periods
FROM `group7-fa25-mgmt58200-final.safegraph.visits` AS v
JOIN `group7-fa25-mgmt58200-final.safegraph.places` AS p
  ON v.safegraph_place_id = p.safegraph_place_id
WHERE p.naics_code = 453210; --- (Office Supplies, Stationery, and Gift Stores)
```

```
-- Identify where office supply stores are most concentrated
```

```
SELECT
  p.region AS state,
  COUNT(DISTINCT p.safegraph_place_id) AS num_stores
FROM `group7-fa25-mgmt58200-final.safegraph.places` AS p
WHERE p.naics_code = 453210
GROUP BY state
ORDER BY num_stores DESC
LIMIT 10;
```

```
-- Get central tendency of store performance
```

```
SELECT
  AVG(v.raw_visit_counts) AS avg_visits_per_record,
  APPROX_QUANTILES(v.raw_visit_counts, 2)[OFFSET(1)] AS median_visits,
  AVG(v.median_dwell) AS avg_dwell_time_minutes,
  APPROX_QUANTILES(v.median_dwell, 2)[OFFSET(1)] AS median_dwell_time
FROM `group7-fa25-mgmt58200-final.safegraph.visits` AS v
JOIN `group7-fa25-mgmt58200-final.safegraph.places` AS p
  ON v.safegraph_place_id = p.safegraph_place_id
WHERE p.naics_code = 453210;
```

```
-- Basic data overview: total number of records and date coverage
```

```
SELECT
  COUNT(*) AS total_rows,
  MIN(v.date_range_start) AS start_date,
  MAX(v.date_range_start) AS end_date
FROM `group7-fa25-mgmt58200-final.safegraph.visits` v
JOIN `group7-fa25-mgmt58200-final.safegraph.places` p
  ON v.safegraph_place_id = p.safegraph_place_id
WHERE p.naics_code = 453210; -- Filter to office supply, stationery, and gift stores
```

```
-- Compute the average daily visits and median dwell time
```

```
SELECT
```

```
    AVG(v.raw_visit_counts) AS avg_daily_visits,  
    AVG(v.median_dwell) AS avg_dwell_minutes  
FROM `group7-fa25-mgmt58200-final.safegraph.visits` v  
JOIN `group7-fa25-mgmt58200-final.safegraph.places` p  
    ON v.safegraph_place_id = p.safegraph_place_id  
WHERE p.naics_code = 453210;
```

```
-- -----  
-- Find which states saw the highest traffic during Jan-Apr 2020  
-- -----
```

```
SELECT  
    p.region AS state,  
    SUM(v.raw_visit_counts) AS total_visits  
FROM `group7-fa25-mgmt58200-final.safegraph.visits` v  
JOIN `group7-fa25-mgmt58200-final.safegraph.places` p  
    ON v.safegraph_place_id = p.safegraph_place_id  
WHERE p.naics_code = 453210  
    AND v.date_range_start BETWEEN '2020-01-01' AND '2020-04-30'  
GROUP BY state  
ORDER BY total_visits DESC  
LIMIT 10;
```

Appendix 2: AI Support Statement

Generative AI was used as a supporting tool throughout the analysis and report creation process for this project. The primary platform utilized was Perplexity AI, with additional use of other generative AI tools as needed for code debugging and writing assistance. All AI-generated content was critically reviewed, revised, and validated by group members for methodological soundness and academic integrity.

How We Used AI

- Generated and refined exploratory SQL suggestions based on our research questions, supporting rapid data exploration in SafeGraph and Census datasets.
- Drafted, edited, and organized project documentation and narrative components of the report, including research background and analysis sections.
- Assisted with troubleshooting errors in SQL code by providing recommended syntax repairs and logic checks.

Example Prompts Used

- "Suggest demographic features from the cbg_demographics dataset relevant for analyzing post-pandemic retail recovery."
- "Summarize preliminary findings about brand-specific differences in visit recovery at office supply stores."
- "Explain what dwell time is and what's working age group and the group corresponding to the education."